Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER B.TECH. (ELECTRICAL &ELECTRONICS)

END SEMESTER EXAMINATIONS, NOVEMBER-DECEMBER 2024

DIGITAL SIGNAL PROCESSING [ELE 3122]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 23 NOVEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Determine convolution sum of $x_1(n)$ and $x_2(n)$ without using the properties of z transform or DTFT where

$x_1(n) = \alpha^n u[n]$ and $x_2(n) = \begin{cases} 2, & 0 \le n \le 4 \\ 0, & otherwise \end{cases}$

Assume $\alpha < 1$ ***(04)***

1B. Justify the discrete time signal $x[n]$ is an energy signal, where

$x(n) = \left(\frac{1}{4}\right)^n u(n)$ ***(02)***

1C. Demonstrate the overlap-add method to compute the linear convolution of the following sequences. $h[n] = [\underset{\uparrow}{1}, 1, 1]$, and

$x[n] = [\underset{\uparrow}{1}, 2, 0, -3, 4, 2, -1, 1, -2, 3, 2, 1].$

Assume sub-frame length =4. ***(04)***

2A. Determine $Y[e^{j\omega}]$ if y(n) =0.5 [x(n+1)-x(n-1)]. Use the properties of DTFT. Plot the magnitude and phase response of this system considering y(n) as output and x(n) as input to the system. ***(03)***

2B. For the sequence $x[n] = [\underset{\uparrow}{2}, 2, 0, 2]$,

(i) Calculate the DFT of the data sequence $x[n]$

(ii) Calculate the DFT of the circularly right shifted $x[n]$ by 2 units.

(iii) Determine the number of complex multiplications & additions required to compute the 4-point DFT? ***(03)***

2C. Apply DIT-FFT radix-2 algorithm, to compute the 8-point DFT of the sequence $x[n] = [\underset{\uparrow}{2}, 1, 1, 2]$. Draw the Butterfly diagram (signal flow graph). Show all the intermediate results to arrive at final answer on the Butterfly diagram. ***(04)***

3A It is required to optimize the signal-to-noise ratio and sensitivity of a signal receiver application satisfying the linear phase constraint. Achieve it by designing a suitable band-pass FIR filter to pass frequencies in the range $1\ to\ 2$ rad/sample by taking 7 samples of rectangular window sequence. Determine the system function and suitable structure to implement the same. Draw the structure. ***(05)***

3B Determine the location of pole and zero of an LTI system described by the difference equation $y[n]-\frac{3}{4}y[n-1]+\frac{1}{8}y[n-2]=x[n]+3x[n-1]+3x[n-2]$. Show the pole and zero location on z plane with Region of Convergence if the system is causal. Check the stability of system. ***(03)***

3C Find the inverse z-transform of $X[z]$ using partial fraction technique.

$X[z]=\frac{z(z-6)}{z^2-6z+8}$; ROC $|z|>4$ ***(02)***

4A Given the sequence $x[n]=u[n]+jv[n]$, and the 6-point DFT of x[n] is $X[k]=[8+j5,\ -5-j6,\ 3+j2,\ 6+j4,\ -4-j2,\ 5+j3]$. Compute the 6-point DFT of $u[n]$ and $v[n]$. ***(03)***

4B Obtain the Direct form-I and Direct form-II structure for the following system $y[n]=\frac{3}{4}y[n-1]-\frac{1}{8}y[n-2]+x[n]+\frac{1}{3}x[n-1]$. Why Direct form II structure is better than Direct form-I structure? ***(03)***

4C Determine the 3-dB bandwidth of the filter $H_1(z)=\frac{1-a}{1-az^{-1}}$.

If $H_2(z)=\frac{1-a}{2}\frac{1+z^{-1}}{1-az^{-1}}$, among $H_1(z)$ and $H_2(z)$ which is better lowpass filter. Justify your answer. Assume a=0.9 ***(04)***

5A Consider a continuous time-domain sinewave defined by

$$x(t)=\cos(4000\pi t)$$

that was sampled to produce the discrete sinewave sequence defined by

$$x[n]=\cos(n\pi/2)$$

What is the *Fs* sample rate, measured in Hz, that would result in sequence $x[n]$? ***(02)***

5B Apply Impulse Invariance Technique to convert analog filter defined by the transfer function $H(s)=\frac{4s+11}{s^2+5s+6}$ to its equivalent digital filter. Assume sampling frequency of 10 Hz. ***(03)***

5C For a certain biomedical application, it is required to remove a band of high frequencies. Design a suitable digital IIR Butterworth filter using Bilinear transformation technique to meet the following specifications:

$$0.8\ \le \left|H\left(e^{j\omega}\right)\right| \le 1\ ;\quad 0\le\omega\ \le 0.3\pi$$

$$\left|H\left(e^{j\omega}\right)\right| \le 0.35\ ;\ \ 0.7\pi\le\omega\le\pi$$

Take T = 1 sec. ***(05)***